

IK 512: Rhythmic Structures and Applications in Music and Musopathology

Course Code	IK 512
Course Name	Rhythmic Structures and Applications in Music and Musopathology
L-T-P-C	3-0-0-3
Prerequisites	None
Intended for	BTech/MTech/MS/MSc/MA/Ph.D
Mutual Exclusion	None
Approval	42nd Senate (Jan 2024)

Course Contents

- **Introduction to Rhythm:** Rhythm as a Fundamental Operating Principle in the Universe and a regulating health mechanism for all living organisms right from heartbeat. (2 Hours)
- **Fundamentals of Rhythm:** Understanding rhythm basics common to world music systems; time signatures; tempo; speed and beat. (2 Hours)
- **Introduction to India's approach to rhythm:** Global popularity of Indian rhythms; Concept of Laya & Tala; Parts of Talas (angas); Finger counting (Jaati and Laghu); Pulse and Gait within beats (Gati/Nadai); Kaala and Kaalapramana; Types of Patterns (Yatis) etc. (6 Hours)
- **Percussion Instruments of India:** Types (Skin based, body based etc); Ancient instruments; prominent contemporary instruments; Drum language in India. (3 Hours)
- **Rhythmic performances:** Percussion accompaniment to melody in Carnatic and Hindustani traditions; percussion interludes and improvisation; percussive cadenzas and climaxes. (3 Hours)
- **Rhythmic Expressions Across Cultures:** Examination of rhythmic complexities in various world music traditions, including Indian classical, African, Latin American, and Western music. (4 Hours)
- **Cognitive Aspects of Rhythm:** Exploring the relationship between rhythm and cognitive functions, including memory, attention, and motor coordination. (4 Hours)
- **Rhythm in Musopathology:** Studying the therapeutic applications of rhythm, its impact on psychological states, Use of Vocal Percussion in speech rehabilitation of stroke victims or patients with other conditions. (6 Hours)
- **Technological Tools for Rhythmic Analysis and Creation:** Introduction to software and digital tools for rhythm analysis, creation, and its applications in music therapy. (4 Hours)
- **Workshop and Case Studies:** Practical workshops on creating rhythmic compositions; case studies on using rhythm in therapeutic settings. (8 Hours)

Laboratory Modules

1. NA

Text Books

1. Sadanand Naimpali, **Theory and Practice of Tabla - The Secular Rationalist Reformer**, Popular Prakashan Private Limited, 2011.
2. Sankaran, T. S., **The Art of Konnakol (Solkattu): Spoken Rhythms of South Indian Music**, Lalith Publishers, 2010.

References

1. Clayton, M., Sager, R., & Will, U., **In Time with the Music: The Concept of Entrainment and Its Significance for Ethnomusicology**, ESEM Counter Point, 2005.
2. Hartenberger, R. (Ed.), **The Cambridge Companion to Percussion**, Cambridge University Press, 2016.
3. London, J., **Hearing in Time: Psychological Aspects of Musical Meter**, 2nd Edition, Oxford University Press, 2012.
4. Sankaran, T. S., **The Rhythmic Principles and Practice of South Indian Drumming**, Lalith Publishers, 1994.
5. Thaut, M. H., **Rhythm, Music, and the Brain: Scientific Foundations and Clinical Applications**, Routledge, 2005.